BYTEBITE PH: SMART HEALTHY MEAL PLANNER & BUDGET LOCATOR APP

SDG Innovation Challenge: From Problem to Sustainable Solution (Lean Canvas, Service Blueprint & Financials)

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Market Sizing: TAM 600,000 | SAM 25,000 (Cubao, QC) | SOM (Quantity Q): 1,000 Active Paid Subscribers / Month

Live Cloud Web App: https://renier-l.github.io/bytebite-ph/ | SDGs: SDG 3, SDG 2, SDG 1, SDG 12

1. TITLE & STEP 1: SDG PROBLEM STATEMENT

- **Primary SDGs Addressed: SDG 3 (Good Health and Well-Being - Preventing Chronic Kidney** Disease/Dialysis, Hypertension & Diabetes), SDG 2 (Zero Hunger), SDG 1 (No Poverty), and SDG 12 (Responsible Consumption).

3-5 Sentence Problem Statement (Answering All 5 Required Questions):

Across urban university corridors in the Philippines such as Cubao, Quezon City, there is a high and rising prevalence of preventable lifestyle diseases - specifically Chronic Kidney Disease (CKD) requiring dialysis, hypertension, and diabetes - among young adults (1. What is the problem? & 3. Where does it occur?). University students and entry-level workers on tight budgets (PHP 45 to PHP 150 per meal) and busy schedules rely heavily on cheap, high-sodium fast food, instant noodles, and canned goods packing 1,500mg-1,900mg of sodium per plate because they lack a tool to plan affordable healthy meals or locate healthy local karinderyas (2. Who is affected?). According to the National Kidney and Transplant Institute (NKTI), over 1 in 10 Filipinos suffer from kidney disease, making diet-induced renal failure both a life-threatening crisis for young people and a catastrophic PHP 240,000+/year financial burden for families (4. Why does it need to be solved?). Addressing this urgent problem requires a pure digital smart meal planner and budget karinderya locator aligned with UN SDG 3, SDG 2, and SDG 1 (5. What SDG does it support?).

2. STEP 2: PRODUCT / SERVICE CONCEPT

- **Product/Service Name: ByteBite PH: Smart Healthy Meal Planner, Kidney Shield & Budget** Karinderya Locator App (Pure Digital Web & Mobile App)
- **Target User & Market Size: University students and young adults in Cubao, Quezon City** (TAM: 600,000 Metro Manila | SAM: 25,000 Cubao, QC | SOM: 1,000 active paid subscribers/month).
- **One-Sentence Description: ByteBite PH is a pure digital cloud application that generates** customized, kidney-safe Filipino meal plans across student budgets (Low PHP 45-75, Medium PHP 80-135, High PHP 140-235), scans meals in real time via AI camera for sodium risk, and locates affordable healthy karinderyas in Cubao, QC.

6-Sentence Explanation of How ByteBite Works:

1. ByteBite PH is a pure digital web and mobile application that empowers busy students to prevent lifestyle diseases (CKD/dialysis, diabetes, hypertension) within their daily food allowance. 2. It generates customized Filipino meal plans across Low (PHP 45-75), Medium (PHP 80-135), and High (PHP 140-235) tiers with itemized palengke ingredient costs and 12 H.264 MP4 audio-narrated Cooking Reels. 3. Its interactive Live GPS Map (Leaflet + OpenStreetMap) instantly locates 6 verified, affordable healthy karinderyas in Cubao, QC with real Haversine distance routing. 4. Its Real-Time AI Camera Scanner (TensorFlow.js MobileNet) identifies meals live and flags high-sodium dialysis-risk foods while recommending PHP 50 low-salt swaps. 5. The Kidney & Sodium Shield tracks daily sodium against DOH limits (<1,500mg-2,000mg) alongside an 8-glass water hydration flush tracker. 6. Priced at PHP 99.00/user using Penetration Pricing with an PHP 80.00 contribution margin (after PHP 19.00 variable cost) and PHP 25,000.00 monthly fixed costs, ByteBite breaks even at 313 subscribers and earns PHP 55,000.00 monthly net profit at our 1,000-user SOM.

3. STEP 3: COMPLETE LEAN CANVAS (ALL 9 BLOCKS)

- **Block 1 - Problem: (1) High prevalence of CKD/dialysis, hypertension, and diabetes among** young adults due to high-salt diets; (2) Students rely on cheap instant noodles and fast food (1,500mg-1,900mg Na/meal) due to tight PHP 45-150 budgets; (3) Lack of visibility for affordable healthy karinderyas in Cubao, QC.
- **Block 2 - Customer Segments: Primary Customers: Budget-conscious university students** (TIP QC, Cubao belt) and young adults (TAM: 600,000 | SAM: 25,000 | SOM: 1,000 paid subscribers/mo). End Users: Daily student meal planners, dorm cooks, PC/Mobile users. Beneficiaries: Local Cubao healthy karinderyas and families protected from dialysis costs.
- **Block 3 - Unique Value Proposition (UVP): "Our product/service helps university students** and young adults in Cubao, QC achieve customized, kidney-safe healthy nutrition within their real PHP 45 to PHP 235 budget by providing a pure digital smart meal planner, real-time AI sodium scanner, and instant GPS locator for affordable healthy karinderyas."
- **Block 4 - Solution: (1) 1,009+ Meal Finder across Low/Med/High PHP tiers + 7-Day Pro** Customizer & Palengke Grocery List; (2) Real-Time AI Camera & Barcode Food Scanner with sodium swap alerts; (3) Kidney & Sodium Shield + 8-glass hydration tracker; (4) Interactive Cubao GPS Map + 12 MP4 Cooking Reels with voice narration.
- **Block 5 - Channels: (1) Pure digital cloud hosting (GitHub Pages:** https://renier-l.github.io/bytebite-ph/ & Vercel) with Mobile QR handoff; (2) Campus partnerships & clinics at TIP QC; (3) Partner karinderya QR standees in Cubao; (4) TikTok/Facebook short-form cooking reels.
- **Block 6 - Revenue Streams (1 MONTH ONLY, BASED ON SOM = 1,000): Monthly Subscription** Price (P) = PHP 99.00/user (or PHP 79 Student Plan). Quantity (Q) based on SOM = 1,000 active paid subscribers. 1-Month Total Revenue = Price x Quantity = PHP 99.00 x 1,000 = PHP 99,000.00 (1 Month).
- **Block 7 - Cost Structure (1 MONTH FIXED & VARIABLE COSTS): Monthly Fixed Cost (FC) = PHP** 25,000.00/month (constant: PHP 12,000 dev salary/retainer, PHP 4,500 cloud hosting, PHP 4,000 dataset maintenance, PHP 3,000 campus marketing, PHP 1,500 QR kits). Variable Cost (VC) = PHP 19.00/user x 1,000 SOM users = PHP 19,000.00/month. Total 1-Month Cost = PHP 44,000.00.
- **Block 8 - Key Metrics: (1) Reaching 313 Break-Even subscribers and 1,000 SOM active** subscribers; (2) Generating PHP 99,000.00 1-month revenue and PHP 55,000.00 1-month net profit; (3) Keeping user daily sodium below 1,500mg-2,000mg; (4) Saving students PHP 60-110/day on meals; (5) Monthly GPS locator redemptions at Cubao karinderyas.
- **Block 9 - Unfair Advantage: Hyper-localized 1,009-meal Cubao Karinderya & Palengke** Nutrition Database mapped to real QC prices and DOH kidney sodium limits, paired with a client-side TensorFlow.js Edge AI architecture that keeps variable cost at just PHP 19.00/user.

4. STEP 4: PRODUCT DESIGN, TECHNICAL SPECS & HORIZONTAL SERVICE BLUEPRINT

- **A. Product Name: ByteBite PH: Smart Healthy Meal Planner, Kidney Shield & Budget Locator** (Responsive PC & Mobile Web App)
- **B & C. Labeled Design & Dimensions: Desktop PC Viewport: 1440x900px widescreen with** fixed 262px navigation sidebar and sticky topbar. Mobile Viewport: 390x844px (fits 71.5mm x 146.7mm smartphones) with 7-tab thumb-zone bottom bar. Capacity: 1,009 Filipino meals, 12 H.264 MP4 cooking videos, 6 GPS Cubao karinderyas, 0g physical weight.
- **D & E. Materials & Main Components: HTML5, CSS3, ES2022 JS, TensorFlow.js 4.17 MobileNet** v2 (Client-Side Edge AI Vision), Leaflet 1.9.4 + OpenStreetMap (GPS Haversine Routing), Web Speech Synthesis API (Voice Narration), GitHub Pages / Vercel Serverless Cloud Engine.
- **F. Safety Features (2+): (1) DOH/WHO Renal Sodium Safety Shield (<1,500mg optimal /** 2,000mg max limit) with automatic high-sodium dialysis risk alerts; (2) On-device browser camera processing for user privacy plus scrypt password hashing.
- **G. Sustainability Features (2+): (1) 100% pure digital zero-hardware footprint** eliminating paper/plastic waste; (2) Shifts student diets from plastic-wrapped instant noodles to fresh local vegetables (monggo, malunggay, kangkong, tokwa) from neighborhood Cubao karinderyas.

H. HORIZONTAL SERVICE BLUEPRINT (CUSTOMER PERSPECTIVE BY LAYER: START TO END)

Flow Direction: Horizontal from Stage 1 (START: Enter Online Web App) -> Stage 2 (Discover Budget Meals) -> Stage 3 (Scan Food Live via AI) -> Stage 4 (Watch Reels / Locate GPS Kitchen) -> Stage 5 (END: Log Meal, Redeem QR & Exit Online Web App).

- **Layer 1 - Physical & Digital Evidence (Visible Proof): [Stage 1: START] Live URL** (renier-l.github.io/bytebite-ph) & QR Code -> [Stage 2] Low/Med/High PHP Meal Cards -> [Stage 3] Live Camera HUD & Sodium Verdict Badge -> [Stage 4] Palengke Cost Table, 12 MP4 Videos & Leaflet Map Pins -> [Stage 5: END] Updated Progress Bars, Printable Grocery List & VIP QR Pass.
- **Layer 2 - Customer Actions (Customer Perspective): [Stage 1: START] Student enters** online web app on PC/Mobile & signs in -> [Stage 2] Selects PHP 45-235 budget tier & Low-Sodium goal -> [Stage 3] Points camera at real food plate to check sodium risk -> [Stage 4] Watches cooking reel w/ audio or clicks Locate My Real GPS -> [Stage 5: END] Logs meal + 8 glasses water, claims VIP voucher & exits web app.
- **Layer 3 - Frontstage / Frontend Actions (Visible UI - app.js): [Stage 1]** syncDeviceViewport() adapts layout to PC or Mobile -> [Stage 2] generator() filters 1,009 meals & renders cards -> [Stage 3] startCameraScan() streams video & shows healthy swap button -> [Stage 4] setupReels() plays H.264 MP4 + voice; initLeafletMap() renders pins -> [Stage 5] trackMeal() updates bars; claimProVoucher() displays QR.

--- LINE OF VISIBILITY (Above is visible on Frontend UI; Below runs in Backend & Support Engine) ---

- **Layer 4 - Backstage / Backend Actions (Hidden Logic - server.js / Cloud Engine): [Stage** 1] /api/bootstrap & getStaticDB() load user profile & goals -> [Stage 2] Multi-filter query ranks meals by price & <=450mg sodium -> [Stage 3] classifyImageElement() runs TensorFlow.js MobileNet v2 + RGB pixel analysis -> [Stage 4] haversineClient() computes exact GPS km to 6 Cubao kitchens -> [Stage 5] /api/tracker/log & saveStaticDB() persist daily totals.
- **Layer 5 - Support Processes & External Entities: [Stage 1] GitHub Pages / Vercel CDN &** SSL encryption -> [Stage 2] public/data/db.json (1,009 PH meals & palengke dataset) -> [Stage 3] TensorFlow.js CDN & Open Food Facts Barcode API -> [Stage 4] OpenStreetMap Tile Servers & Web Speech API -> [Stage 5] GCash/Maya Gateway & 6 Cubao Partner Karinderyas.

5. STEP 5: 1-MONTH SOM PROFIT COMPUTATION, PRICING STRATEGY & COST CONTROL

A. Basis of Quantity (Q) from TAM, SAM, SOM (1 Month Only):

- **Market Sizing Basis: TAM = 600,000 (Metro Manila) | SAM = 25,000 (Cubao, QC) | SOM =** 1,000 active paid subscribers/month. Following the rule that SOM is the exact basis for Quantity (Q), our 1-Month Quantity is Q = 1,000 subscribers.

B. 1-Month Revenue Stream (Revenue = Price x Quantity):

- **1-Month Total Revenue: Selling Price (P) = PHP 99.00/user/month x Quantity (Q_SOM) =** 1,000 subscribers = PHP 99,000.00 Total Revenue (for 1 Month).

C. 1-Month Cost Structure (Fixed Cost vs. Variable Cost):

- **1. Fixed Costs (Constant at PHP 25,000.00 / month): Labor/Developer Retainer = PHP** 12,000.00 | Manufacturing/Cloud Hosting & SSL = PHP 4,500.00 | Materials/Dataset Maintenance = PHP 4,000.00 | Transportation & Campus Marketing = PHP 3,000.00 | Packaging/Digital Onboarding & QR Kits = PHP 1,500.00. Total Fixed Cost (FC) = PHP 25,000.00/month.
- **2. Variable Costs (Increases with Quantity Q: PHP 19.00/user x 1,000 SOM = PHP 19,000.00** / month): Cloud API & database sync queries (PHP 8.00/user) + SMS/email hydration & sodium alerts (PHP 6.00/user) + GCash/Maya payment gateway fee (PHP 5.00/user) = PHP 19.00/user. At Q = 1,000 SOM subscribers, Total Variable Cost (VC) = PHP 19,000.00/month.
- **3. Total 1-Month Cost (TC = FC + VC): PHP 25,000.00 Fixed Cost + PHP 19,000.00 Variable** Cost = PHP 44,000.00 Total Monthly Cost (or PHP 44.00 Total Cost per Unit at 1,000 users).

D. 1-Month Profit Computation & Break-Even Analysis:

- **Profit Formula (Profit = Total Revenue - Total Cost): PHP 99,000.00 (1-Month Revenue) -** PHP 44,000.00 (1-Month Total Cost) = PHP 55,000.00 Projected Net Profit (for 1 Month).
- **Per-Unit Profit & Contribution Margin: Selling Price (PHP 99.00) - Variable Cost (PHP** 19.00) = PHP 80.00 Contribution Margin/user (80.8%). Net Profit per Unit at 1,000 SOM = PHP 99.00 - PHP 44.00 = PHP 55.00 profit/user. Break-Even Volume (BEP) = PHP 25,000 Fixed Cost / PHP 80 Contribution Margin = 313 subscribers/month.

E. Pricing Strategy Justification (How It Became PHP 99.00):

- **Penetration Pricing Strategy + Cost-Based Floor: At 1,000 SOM users, our total cost per** unit is PHP 44.00. We use Penetration Pricing at PHP 99.00/month (only PHP 3.30/day, psychologically under PHP 100) to undercut foreign nutrition apps (PHP 499-990/mo) by 80% and rapidly capture price-sensitive Cubao students while earning a 125% markup over total unit cost.

F. How to Lower the Cost & Maximize Profit:

- **Cost Reduction & Profit Maximization: (1) Client-Side Edge AI (TensorFlow.js runs in the** user browser) cuts cloud GPU variable costs by 60%; (2) Open-source OpenStreetMap + GitHub Pages/Vercel static hosting eliminates map/server licensing fees; (3) Partner karinderya QR standees provide zero-cost organic marketing; (4) Every subscriber above 313 adds PHP 80.00 pure profit.

6. STEP 6: SDG IMPACT (SOCIAL, ENVIRONMENTAL & ECONOMIC VALUE)

- **Social Impact (SDG 3 & SDG 2): Prevents Chronic Kidney Disease (dialysis), hypertension,** and diabetes among young Filipino adults by replacing salty instant noodles with kidney-safe (<450mg Na/meal) Filipino dishes and hydration tracking.
- **Environmental Impact (SDG 12 & SDG 13): 100% pure digital app that shifts campus dining** away from single-use plastic noodle sachets and fast-food packaging toward fresh wet-market vegetables cooked by local karinderyas.
- **Economic Impact (SDG 1 & SDG 8): Saves students PHP 1,500-2,500/month on food, shields** families from PHP 240,000+/year dialysis bills, drives daily customer foot traffic to 6 Cubao partner karinderyas, and generates PHP 55,000/month net profit.

7. SHORT REFLECTION & REFERENCES

- **1. What I learned: I learned that solving an SDG health crisis like kidney disease among** students requires aligning nutritional science with real Philippine student economics (PHP 45-235 budgets) and sustainable unit economics.
- **2. Most difficult part of the Lean Canvas: Balancing a student-friendly Penetration** Price (PHP 99/mo) against monthly fixed and variable costs so that our 1,000-user SOM achieves a strong PHP 55,000 one-month net profit.
- **3. Future improvements: Integrate direct POS inventory syncing with Cubao karinderyas** and expand our verified low-sodium partner kitchen network across U-Belt Manila and Katipunan.
- **References: (1) National Kidney and Transplant Institute (NKTI) / DOH Philippine Renal** Disease Registry; (2) FNRI-DOST Philippine Food Composition Tables; (3) UN Sustainable Development Goals (SDGs 1, 2, 3, 12); (4) OpenStreetMap & Open Food Facts Nutrition Database.